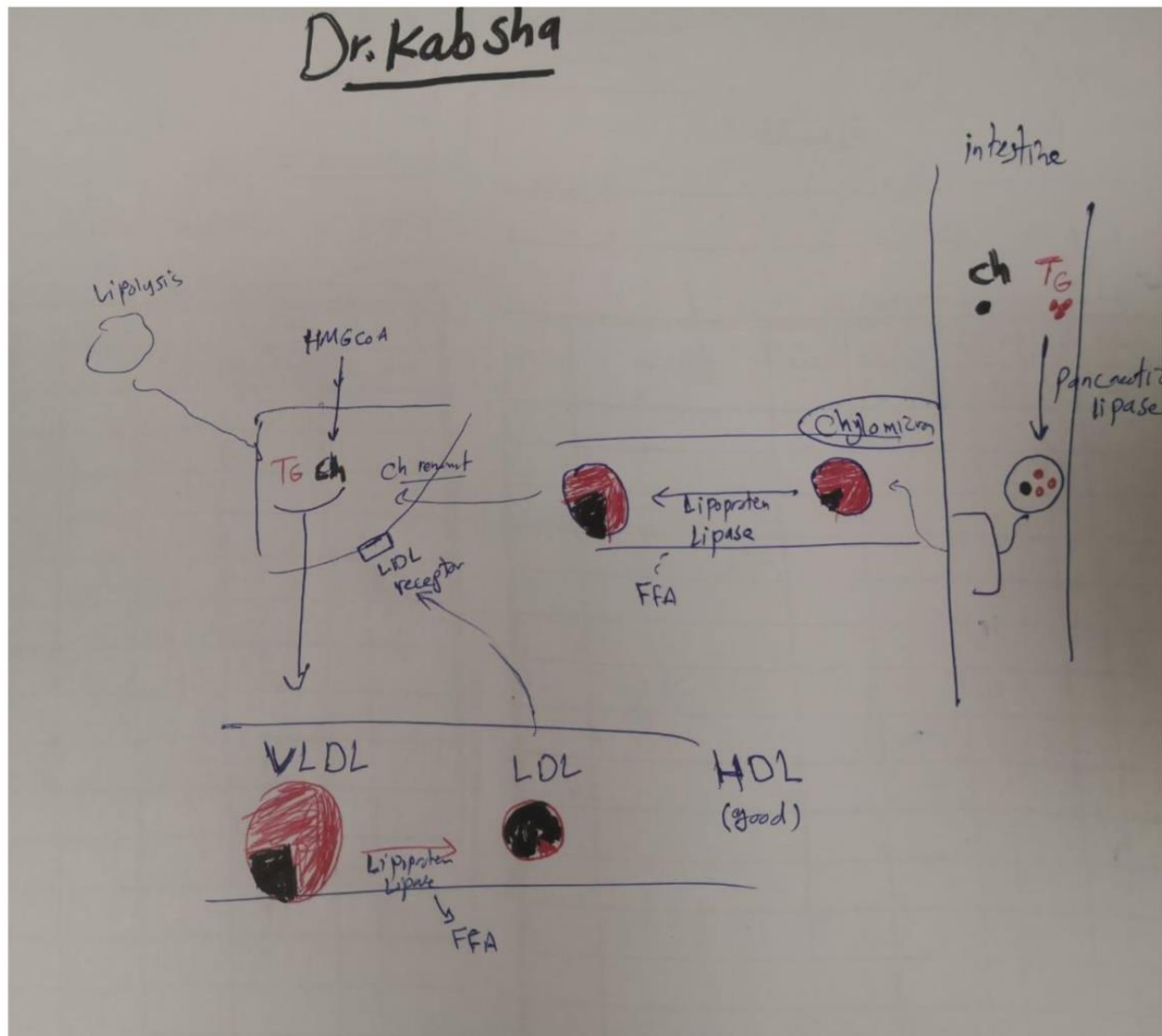
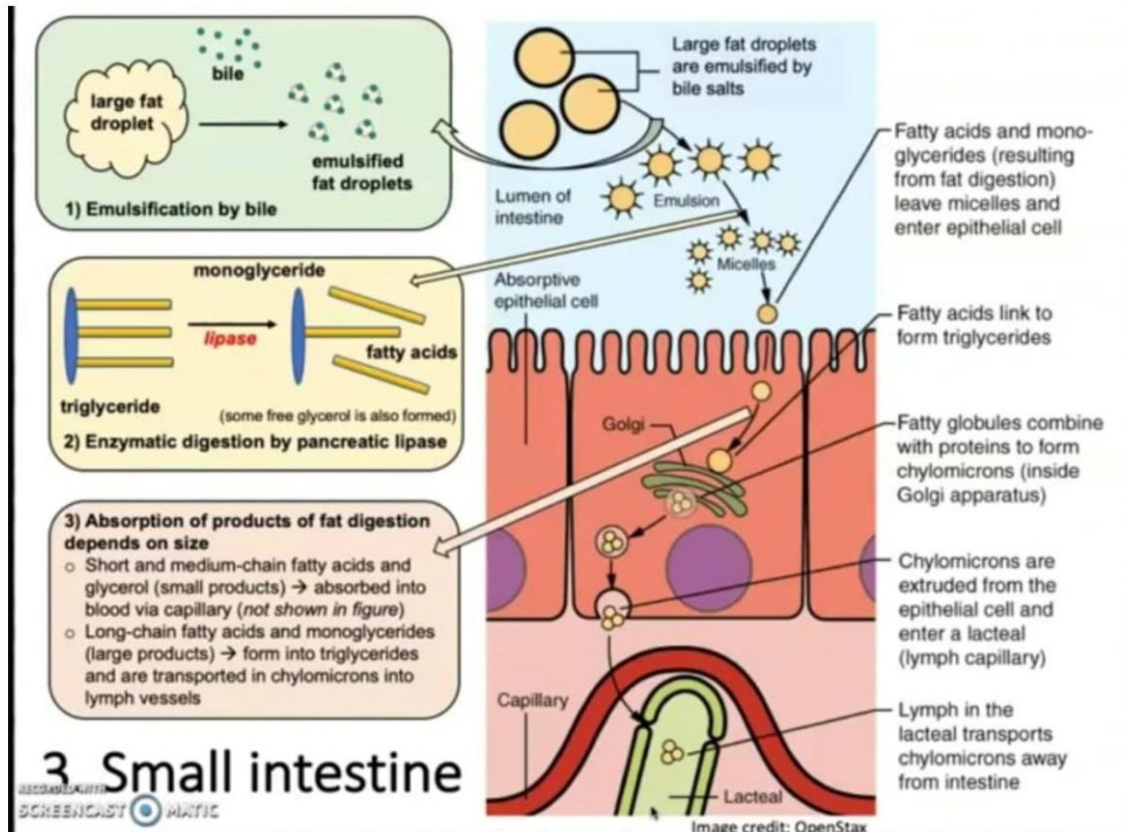


Dyslipidemia

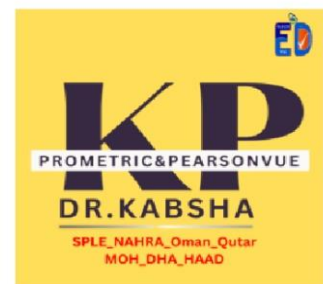


- Cholesterol exits the body either as **free cholesterol** or as **bile acid**.



↑
ch

↑
TG



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- Drug affect lipid profile?

Hypothyroidism → hyperlipidemia

- **Limit** intake of **saturated** fat, **trans** fat (partially hydrogenated) and cholesterol

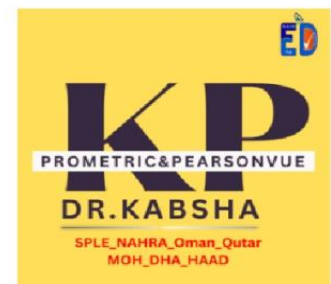
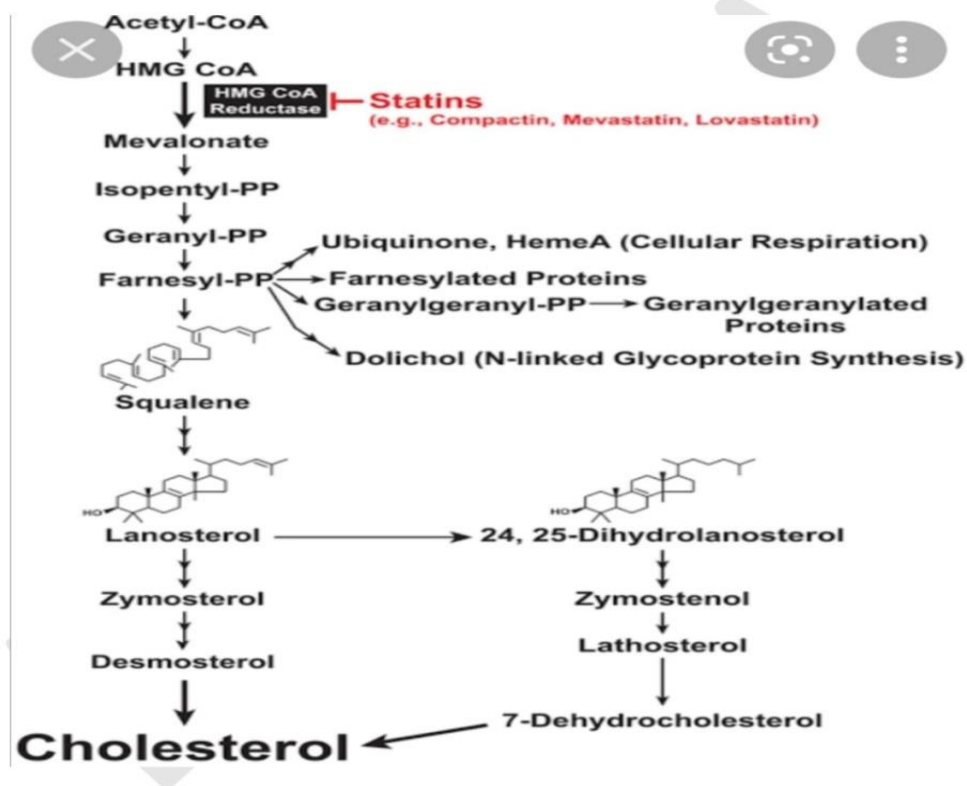
Medication :

- Statin
 - **Bile acid Sequestrants:**(Resins)
 - **Fibric acid derivative:**(Fibrates)
 - Niacin (Vit B3)
 - Cholesterol absorption inhibitors (Ezetimibe).# Can use it with statin
 - Omega – 3 – fatty acid
-
- Chylomicrons are large triglyceride-rich lipoproteins function to transport dietary lipids and cholesterol from intestine to the peripheral tissues.

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❖ **Statin**

MOA: competitively blocking the active site of the first and key rate-limiting enzyme in the mevalonate pathway, HMG-CoA reductase
HMG-CoA reductase inhibitor



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➡ **Not use**

In **pregnant** and **lactating** woman (Categor**X**)??....warfarin..methotrexate..finastride

➡ **SE**

Muscle pain (rhabdomyolysis, myopathy), **Elevate liver enzyme**, **exaggerated glucose level**

➡ **Monitor**

Creatine Kinase, **liver enzyme** (if increases **X 3 fold** from upper limità Hepatotoxicà **DC**)

➡ **Decrease myopathy**

With statin co-administration with **Co-enzyme Q10**

➡ Statin + Gemfibrozil or Valproic acid à **severe Rhabdomyolysis**

❖ **Administration of Statin**

SRP

Statins **need dose** adjustment: **S**imvastatin, **R**osuvastatin, **P**ravastatin

AF

Statin **NOT** need adjustment: **A**torvastatin, **F**lucastatin

FAR is statin ☀ / □ ➡ **F**lucastatin Extended release, **A**torvastatin, **R**osuvastatin

SPFL is statin short acting: ➡ **S**imvastatin, **P**ravastatin, **F**lucastatin, **L**ovastatin

FPRP are statin safe when taken with **grape fruit juice** :**F**lucastatin, **P**itovastatin, **R**osuvastatin, **P**ravastatin

ASL not give with grape fruit

PFR hydrophilic statin

Bile acid Sequestrants:(Resins)

Cholestyramine, **Colesevelam** (option in a pregnant), **Colestipol**

(increase TG),,,, **CI in Hypertriglyceridemia ≥ 300**

take all other drugs -least 1 - 4 hours before or 4 - 6 hours after the bile acid sequestrants.

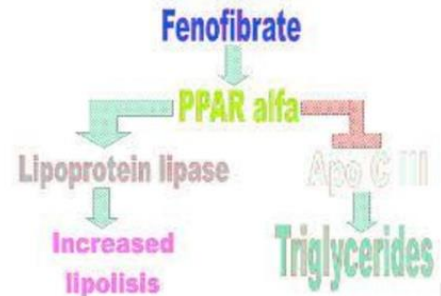
Fibric acid derivative:(Fibrates)

Fenofibrate, **gemfibrozil**

✚ Not use with statin à severe Rhabdomyolysis

✚ Fibrates such as fenofibrate **activate peroxisome proliferator-activated receptors alpha (PPAR-alpha)**, which upregulate lipoprotein lipase, induce high-density lipoprotein (HDL) synthesis, and decrease liver production of apolipoprotein C

- ➡ **Act on triglycerides and cholesterol:** Simvastatin
- ➡ **Acts on triglycerides:** gemfibrozil



- Niacin (Vit B3)
- Cholesterol absorption inhibitors (Ezetimibe). # Can use it with statin
- Omega – 3 – fatty acid..... ↑ bleeding risk .
- ↓ TG ➡ turmeric & garlic.

S1co1b1 gene ??????

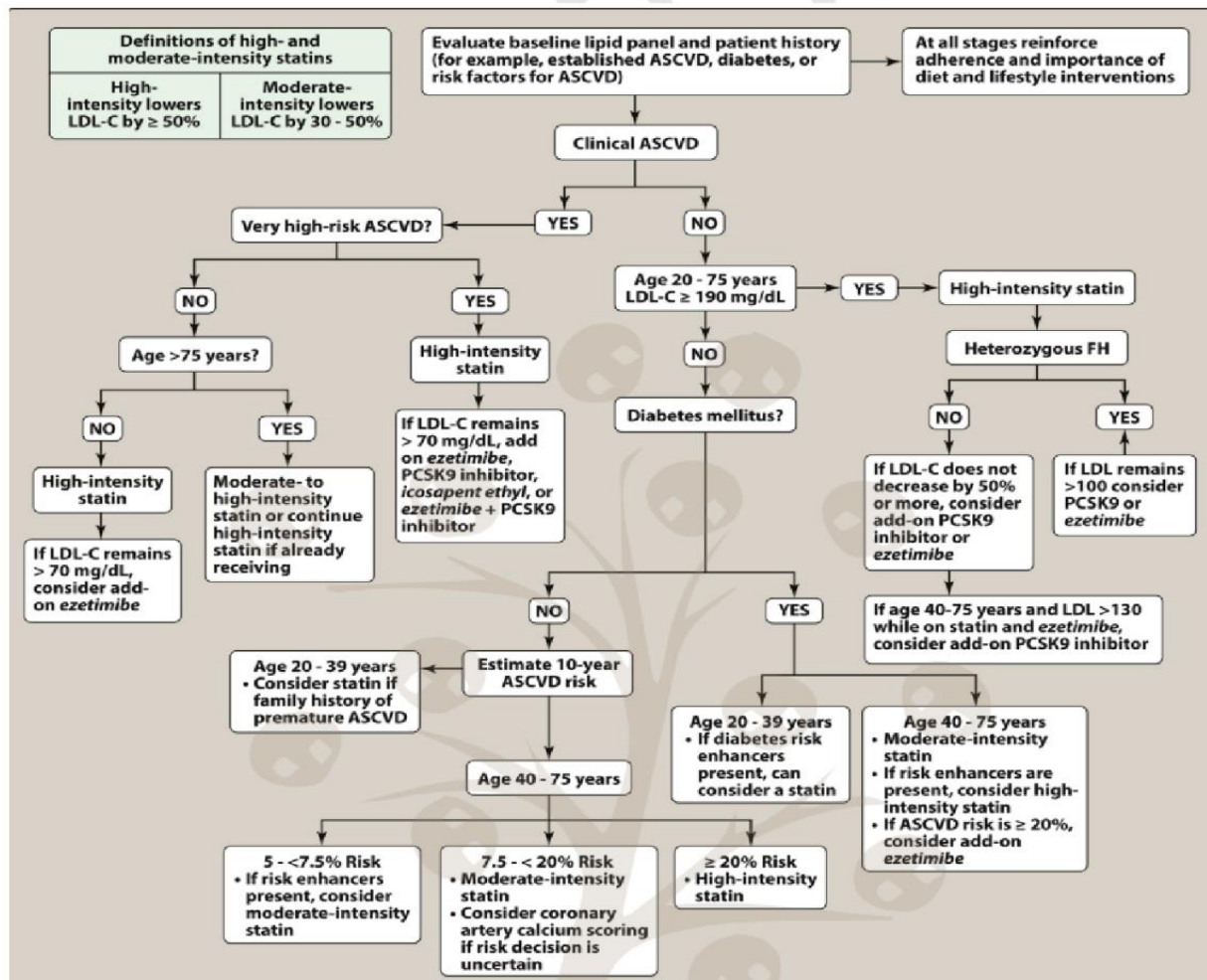
Statin Treatment Intensity Definitions and Selection Options

HIGH-INTENSITY	MODERATE-INTENSITY	LOW-INTENSITY
Daily dose ↓ LDL ≥ 50% Atorvastatin 40-80 mg daily Rosuvastatin 20-40 mg daily	Daily dose ↓ LDL 30% – 49% Atorvastatin 10-20 mg daily Rosuvastatin 5-10 mg daily Simvastatin 20-40 mg daily Pravastatin 40-80 mg daily Lovastatin 40 mg daily Fluvastatin XL 80 mg daily Fluvastatin 40 mg BID Pitavastatin 2-4 mg daily	Daily dose ↓ LDL < 30% Simvastatin 10 mg daily Pravastatin 10-20 mg daily Lovastatin 20 mg daily Fluvastatin 20-40 mg daily Pitavastatin 1 mg daily

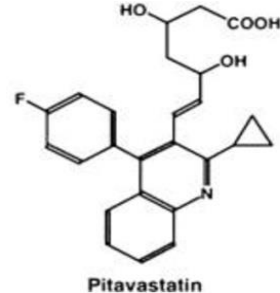
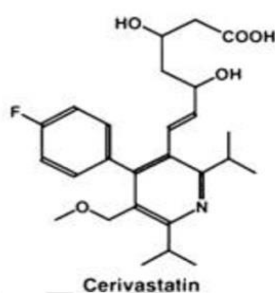
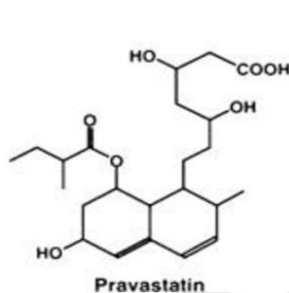
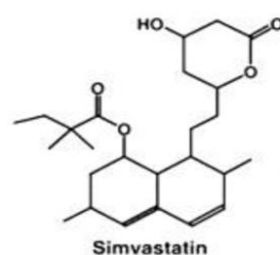
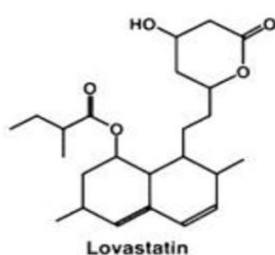
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High intensity	moderate intensity	low intensity
Rosu 20-40mg	Rosu 5-10 mg	sim 10mg
Ator 40-80	Ator 10-20 mg	prava 10-20mg
	Sim 20-40 mg	
	Prava 40-80mg	

✚ Atorva 40mg = Rovista 20mg



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- Pra **natural active**
- Lova **natural prodrug**

طريقتي لحفظ اهم النقاط لادوية الستاتين /

Short-acting statins are more effective when taken in the evening: (SPFL)
السب قل 🌙 يعني بالليل وفيك نوم وتعبان اكيد بتسب العالم

- Simvastatin
- Pravastatin
- Fluvastatin
- Lovastatin

Long-acting statins can be taken in the morning or evening: (FAR)
"فار" 🌞 تلقاه بالليل او بالنهار ماله وقت محدد

- Fluvastatin (extended-release)
- Atorvastatin
- Rosuvastatin

Statin need dose adjustment (SRP) (سرب)
نقول لازم نضبط الجرعه عشان مايصير تسريب 🌊🌊🌊
Sim + Rous+prava

Hydrophilic statin : (PFR)
حفظتها على شكل "بقر" ينحط مع المويا ويحب المويا 🍓💧

statin not give with gripe fruit (ASL)
حفظتها ان هذا اختصار لاسم اصاله 🍌 وهي ماتحب ال gripe

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Which of the following best describes the role of colipase in digestion of lipids?

- A)It breaks down triglycerides into absorbable free fatty acids
- B)It hydrolyzes cholesteryl esters into cholesterol and free fatty acids
- C)It acts as co-enzyme for the action of lipase in the presence of bile acids
- D)It plays an important role in the degradation and excretion of phospholipids

Essential amino acids

I	Isoleucine
Love	Lysine
Lucy	Leucine
Very	Valine
Much	Methionine
Please	Phenylalanine
Try	Tryptophan
To	Threonine
Help	Histidine
Arginine	Arginine

- M A L C **cysteine, Lysine, Arginine and Methionine.** →→ Present in hair and nail. Produce keratine
- BCAA amino acids(**leucine** , isoleucine, valine) decrease **depression & anxiety**

Pepsin act on?

- a)Serine
- b)Lysie
- c)Phenyl alanine
- d)Leucine

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Pepsin on

Cyclic aminoacids tryptophan phenylalanine tyrosine

Monosaccharides: Glucose Galactose Fructose

Disaccharides : Sucrose Maltose Lactose

Polysaccharides : Starch Glycogen Cellulose

Calories

1g fat = 9 calories

1g protein = 4 calories

1g carb = 4 calories

10 g of fat = ???

Golgi apparatus: # Membrane bound organelle found in mast cell. #

Responsible for packing proteins and lipids into vesicles to secretion and therefor plays a key role in the secretory pathway

starch is digested in salive by?

A- Amylase

B- lipase

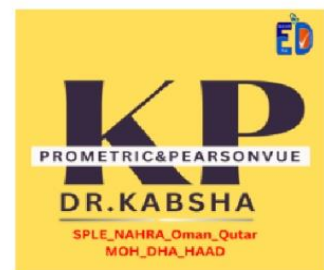
:Omega-6 fatty acid is present in:

a) ☐ Evening primrose

b) Ginseng

C) Aloe

d) Flaxseed



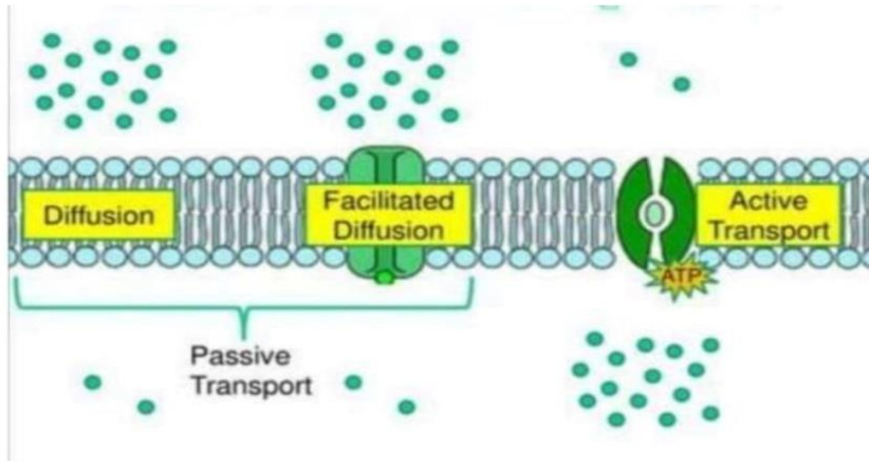
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Which of the following food supplement is rich in omega3 fatty acids?

- A. fish oil**
- B. palm oil**
- C. peanut oil**
- D. coconut oi**

Also:

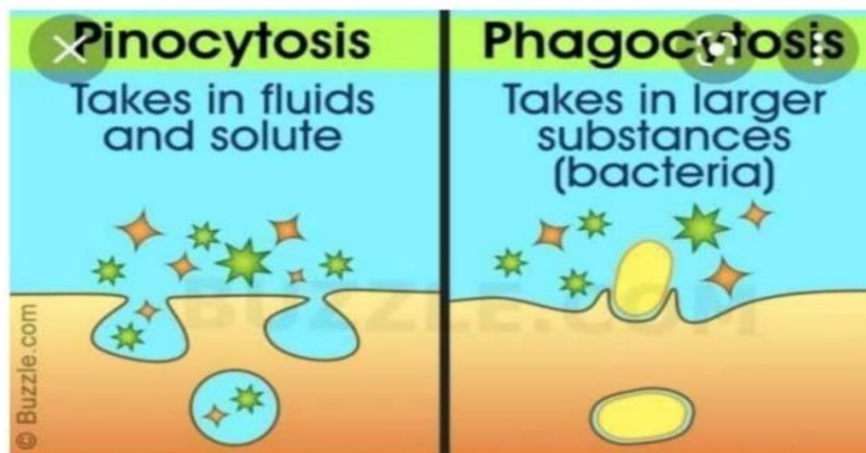
Seaweed and algea good source for omega 3 in vegeterian diet



: Passive with concentration gradient from high to low concentration.. No need for energy

Facilitated diffusion.. Form high to low but Facilitated by transporter

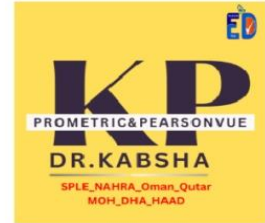
: Active transport.. Need energy.. Against conc gradient.. From low conc to high



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